Journal of Affective Disorders Reports 12 (2023) 100518

Contents lists available at ScienceDirect

Journal of Affective Disorders Reports

journal homepage: www.sciencedirect.com/journal/journal-of-affective-disorders-reports

Research Paper

Heart rate change as a predictor of treatment outcome for ring-coil accelerated low frequency repetitive transcranial magnetic stimulation in major depressive disorder: An exploratory study

Jack Z. Sheen [a,b], Jean-Philippe Miron [a,c,g,h], Helena Voetterl [i,j], Farrokh Mansouri [a], Ryan Zhou [b], Thomas Russell [b], Molly Hyde [a,b], Linsay Fox [b], Zafiris J. Daskalakis [a,c,d,f], John D. Griffiths [a,c,e], Daniel M. Blumberger [a,c,d], Jonathan Downar [a,b,c,*]

[a] Institute of Medical Science, University of Toronto, Toronto, Canada
[b] Krembil Research Institute University Health Network, Toronto, Canada
[c] Department of Psychiatry Faculty of Medicine, University of Toronto, Canada
[d] Temerty Centre for Therapeutic Brain Intervention, Centre for Addiction and Mental Health, Canada
[e] Krembil Centre for Neuroinformatics, Centre for Addiction and Mental Health, Toronto, Canada
[f] Department of Psychiatry, University of California San Diego, San Diego, CA, USA
[g] Centre Hospitalier de l'Université de Montréal (CHUM), Centre de Recherche du CHUM (CRCHUM), Canada
[h] Département de Psychiatrie Faculté de Médecine, Université de Montréal, Montréal, QC, Canada
[i] Brainclinics Foundation Research Institute Brainclinics, Nijmegen, the Netherlands
[j] Department of Cognitive Neuroscience, Faculty of Psychology and Neuroscience Maastricht University, Maastricht, the Netherlands

ARTICLE INFO

Keywords:
Major depressive disorder
Low frequency rTMS
Parabolic rTMS coil
Electrocardiogram
Biomarkers

ABSTRACT

Background and objective: High frequency (HF) left dorsolateral prefrontal cortex (L-DLPFC) rTMS has been reported to induce mild, transient bradycardia in patients with major depressive disorder (MDD), measured using electrocardiography (ECG). Low frequency (LF) rTMS has distinct advantages compared to HF rTMS and our previous investigation of a similar design suggested that heart rate (HR) may have biomarker potential for LF rTMS in MDD patients. Our previous study supported the utilization of HR biomarkers for LF-rTMS in MDD and thus this exploratory replication study aims to further investigate, in a different cohort and with a different coil, the effect of 1Hz right hemisphere (*R*)-DLPFC rTMS on the HR of MDD patients, as well as the potential of using HR as a simple, scalable biomarker to predict rTMS treatment response.

Methods: 24 participants underwent 30 sessions of accelerated 1Hz R-DLPFC rTMS within 7 days after the baseline ECG session, followed by 20–25 once-a-day sessions 7 days after. The Beck Depression Inventory-II score was used as the primary outcome measure.

Results: HR significantly decreased during the first 3 min of the rTMS period. Resting HR, HR during rTMS, and the degree of HR reduction were not significantly associated with treatment outcome.

Conclusion: The results of the current study remain in concordance in trend with that of our previous study despite the use of a less accurate rTMS coil. Therefore, this study further supports the biomarker potential of HR for LF-rTMS in MDD patients.

1. Introduction

Major depressive disorder (MDD) is now the foremost cause of disability worldwide (Friedrich, 2017). As a safe (Rossi et al., 2009) and effective (Brunoni et al., 2017; Lefaucheur et al., 2020; Milev et al., 2016) non-invasive clinical treatment for medication-resistant MDD, repetitive transcranial magnetic stimulation (rTMS) has entered increasingly widespread clinical use. Current objectives of translational rTMS research include the identification of reliable predictors of clinical response to rTMS prior to the start of a treatment course. Predictive

* Corresponding author at: Department of Psychiatry and Institute of Medical Science, Faculty of Medicine, University of Toronto, 27 King's College Circle Toronto, Ontario M5S 1A1 Canada.
E-mail address: jonathan.downar@utoronto.ca (J. Downar).

https://doi.org/10.1016/j.jadr.2023.100518
Received 13 December 2022; Received in revised form 6 March 2023; Accepted 8 March 2023
Available online 14 March 2023
2666-9153/© 2023 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

biomarkers, if proven reliable, could help to identify likely rTMS responders and non-responders, and could also provide objective measures for monitoring engagement of target circuits during treatment. Although clinical features have shown some group-level predictive power, it has been difficult to demonstrate reliable individual-level predictive power for any biomarker to date (Simon and Perlis, 2010; Bagby et al., 2002). The ideal biomarker would, in addition to having high individual-level predictive accuracy, be cost-effective and simple to acquire in real-world clinical settings.

Electrocardiography (ECG) is a convenient and low-cost technique used to measure cardiac activity via recordings of cardiac electrical impulses conducted through the skin, and due to the association between psychiatric disorders and cardiovascular abnormalities, ECG has emerged as a promising biomarker candidate of rTMS treatment outcome. Recent studies have shown that high frequency (HF) rTMS can indeed reduce HR in the general population (Makovac et al., 2017), as well as in MDD patients (Michael and Kaur, 2021; Udupa et al., 2007; Udupa et al., 2011; Iseger et al., 2020). Iseger et al. (2020) also reported a trend towards correlation between HF-rTMS induced HR deceleration and treatment outcome (Iseger et al., 2020). Low frequency (LF) rTMS, which has many safety and tolerability advantages over HF-rTMS (Miron et al., 2020), has also been shown to reduce HR in autism spectrum disorder patients (Casanova et al., 2014). Our previous study has demonstrated that LF-rTMS can reduce HR specifically in MDD patients, and that the degree of reduction was mildly associated with treatment outcome (Sheen et al., 2022). In terms of heart rate variability parameters, one of our previous investigations suggested that the length of our recording sessions, which we did not have an opportunity to adjust for the present study, may have been inadequate for the evaluation of several HRV parameters (Sheen et al., 2022), and another study from our group was unable to find any statistically significant association between the root mean square of the successive differences (RMSSD) and 1Hz rTMS treatment outcome in MDD (JP Miron et al., 2021). Thus, HR has been selected as the focus of the present study.

The specific target region of rTMS may further affect its impact on HR. A 2017 meta-analysis that included various targets of rTMS concluded that the effects of rTMS on HR was most pronounced for rTMS targeting the prefrontal cortex (Makovac et al., 2017). Compared to the left prefrontal cortex, the right prefrontal cortex has been proposed to exert a more inhibitory effect on cardiac output (Ahern et al., 2001). Within the prefrontal cortex, influence on HR may also vary. Kaur et al. (2020) showed that the pattern of rTMS induced HR change was statistically significantly different between the target sites F3 and FC3. Although this study was conducted on healthy participants, the results demonstrated that even small differences in the rTMS target region can potentially influence rTMS induced HR changes (Kaur et al., 2020). Thus, currently available evidence suggests DLPFC rTMS can influence HR, a relationship may exist between DLPFC rTMS-induced changes in HR and clinical response to rTMS treatment in MDD, and the specific target region of rTMS may influence the extent of rTMS induced changes in HR.

Here we present the results of a study examining changes in heart rate elicited by therapeutic stimulation of the right DLPFC at 1Hz with a ring-shaped coil centered over the target region, rather than the more commonly used figure-8 coil. The ring-shaped coil generates a field that is stronger in the area surrounding the DLPFC target and milder on the target itself. This coil was used in this study because it is a significantly lighter coil that can be held more securely by the coil holding arm and thus made coil placement more efficient. Clinically, this approach yielded a less potent therapeutic effect than is typically observed with a figure-8 coil targeting the same region (Miron et al., 2021). Since some individuals still showed response and remission using the ring coil, we hypothesized that 1Hz rTMS delivered by ring coil would reduce HR during the first few minutes of the stimulation period in these individuals, and the degree of reduction would show a negative relationship to treatment outcome; however, such effects and relationships are expected to be weaker in the current study due to the subpar clinical outcome of the ring-coil If so, the findings would support the previous result demonstrating that 1Hz right DLPFC-rTMS would reduce heart rate (Sheen et al., 2022), similarly to high-frequency left DLPFC-rTMS (Iseger et al., 2020). Additionally, if similar trends exist between HR and treatment outcome despite of the utilization of the ring coil, such findings would support our previous result that the degree of reduction may be useful in predicting treatment outcome (Sheen et al., 2022). In another earlier study from our group, for a different cohort of the dataset used in the present study, we briefly reported on the effect of 1Hz R-DLPFC rTMS on HR and the association between HR and treatment outcome (Voetterl et al., 2021). However, this previous analysis was only carried out on HR data recorded during the 1Hz rTMS inter-train-intervals (ITIs): lengthy 30 s resting periods after each train of rTMS pulses during which no rTMS pulses took place. Since the initial sham-controlled investigation of cardiovascular biomarkers of rTMS was conducted using ECG data collected during the stimulation periods and not during the inter-train-intervals (Iseger et al., 2020), as well as considering the possibility that any effect of rTMS on HR may significantly diminish during the ITIs, our group decided to further investigate and report on the ECG data recorded specifically during the 1Hz stimulation periods.

2. Methods

2.1. Study design and participants

24 MDD participants, 13 female and 11 male, with a median age of 41.3, were included in this single-arm, prospective, open-label feasibility study. The sample size of this study was determined based on previous studies of similar purpose and design: Udupa et al. 2007 with $n = 27$ in the rTMS group; Udupa et al. 2011 with $n = 30$ in the rTMS group; Casanova et al. 2014 with $n = 18$; and Iseger et al. 2020 with $n = 15$. Participants were included if they: (1) met criteria for MDD diagnosis (single or recurrent episode) on the Mini International Neuropsychiatric Interview (MINI); (2) maintained a stable medication regimen from 4 weeks prior to the start of the study and could maintain a stable medication throughout the entire study period. Exclusion criteria were: (1) within the past 3 months, having a history of substance dependence or abuse; (2) having concomitant major and unstable medical illness; (3) having a cardiac pacemaker or implanted medication pump; (4) active suicidal intent; (5) as assessed by a study investigator, having a diagnosis of any primary personality disorder that is causing greater impairment than MDD; (6) diagnosis of any psychotic disorder; (7) any significant neurological disorder or insult (including, but not limited to: any condition likely to be associated with increased intracranial pressure, space-occupying brain lesion, any history of seizure confirmed diagnostically by neurological assessment (except those therapeutically induced by ECT), cerebral aneurysm, Parkinson's disease, Huntington's chorea, dementia, stroke, neurologically confirmed diagnosis of traumatic brain injury, or multiple sclerosis); (8) if participating in psychotherapy, must have been in stable treatment for at least 3 months prior to entry into the study (with no anticipation of change in the frequency of therapeutic sessions, or the therapeutic focus over the duration of the study); (9) in the opinion of the investigator, any clinically significant laboratory abnormality; (10) due to the potential to limit rTMS efficacy, a dose of more than lorazepam 2 mg daily (or equivalent) currently (or in the last 4 weeks) or any dose of an anticonvulsant; (11) any non-correctable clinically significant sensory impairment, and (12) any significant cardiovascular or metabolic disorder or insult including, but not limited to: coronary artery disease, abnormal heart rhythms, heart failure, cardiac valve disease, congenital heart disease, cardiomyopathy, vascular disease, dyslipidemia, diabetes, or hypertension. All participants provided informed consent and this study was approved by the Research Ethics Board of the University Health Network hospital in Toronto, Canada.

2.2. rTMS procedures and clinical outcome measure

rTMS was delivered with a MagPro R20 stimulator equipped with a MMC-140 coil (MagVenture, Farum, Denmark). Resting motor threshold (rMT) was determined according to standard techniques (McClintock et al., 2017). Treatment consisted of an accelerated rTMS (arTMS) course of 6 sessions per day (with a minimum inter-session interval of 45 min) over the 5 consecutive weekdays from Monday to Friday, followed by 20 to 25 once-a-day sessions 7 days after the accelerated sessions, totalling 50 to 55 sessions per patient. Each rTMS session consisted of low-frequency 1Hz stimulation delivered over a 8.5-minute period (60 s on and 30 s off; 6 trains in total) at 120% of rMT over the right dorsolateral prefrontal cortex (R-DLPFC). DLPFC was located according to a previously published heuristic approximating the F4 EEG site using scalp measurements (Mir-Moghtadaei et al., 2017).

The Beck Depression Inventory-II (BDI-II) score assessed at the second follow-up, which occurred within 7 days after the end of the tapering sessions, was used as the primary outcome measure. This second follow-up was chosen for comparative purposes with our previous study's primary outcome, as both primary outcomes were assessed at approximately 1 month after the end of the arTMS sessions.

2.3. Cardiac data acquisition

ECG acquisition was performed at the baseline session, which took place within one week prior to the start of the arTMS sessions. As part of a 16-lead, 512 Hz EEG system (EasyCap, Brain Products, Munich, Germany), one electrode was attached to the left wrist of the participants to acquire ECG data. The FCz and AFz electrodes on the EEG cap were used as reference and ground electrodes, respectively. Stable impedances of $<5k\Omega$ were acquired prior to the start of the recording session. Participants first underwent a 5 min. resting state session with no stimulation, then a 1Hz rTMS session (60 s on and 30 s off; 6 trains; 8.5 min total stimulation time; 360 pulses/sessions; 120% rMT), followed by another 5 min resting state session. All recordings were conducted from 9 am to 12 pm during the day. This study focused specifically on the ECG data; results from the EEG recording component are to be the subject of a separate report.

2.4. Cardiac data processing

Raw ECG data were visualized using the Python-based Plotly package (Plotly Technologies Inc 2015). R-peaks of the QRS complex were visually detected and the time at which the peaks occurred were recorded to calculate the RR-interval (time lapse between two consecutive R-peaks), from which the HR was calculated in beats per minute. To avoid artefacts, 3 min of data from the end of the resting state recording were used to calculate resting HR, all of the 6 active trains during the rTMS-EEG/ECG session was used to calculate HR during 1Hz stimulation, and 3 min from the start of the post-treatment resting period recording were used to calculate HR after 1Hz stimulation.

2.5. Statistical analysis

Due to the within-subjects design of the study, linear mixed effects models (LMMs) were conducted using the R (version 4.0.4) (R Core Team 2021) package LME4 (Bates et al., 2015) to test for statistically significant differences in mean HR between the pre-stimulation resting period and the 1Hz rTMS period, as well as between the pre-stimulation resting period and the post-stimulation resting period. The 1Hz rTMS period was divided into six 1-minute segments for the analysis, with each segment representing one stimulation train, since it was previously reported that theta burst rTMS exerted a significant effect on HR during the first minute of stimulation (Iseger et al., 2020). Instead of averaging RR-intervals in each of the Sessions (pre-stimulation resting period, 1Hz rTMS period, and post-stimulation resting period), every RR-interval recorded in a Session were included in the models to more accurately account for the variance structure of the RR-intervals within each subject. The within-subject correlation of HR across the Sessions was accounted for by introducing random intercepts for each subject. The main fixed effects were defined as HR and Session, with every time-point of the RR-interval nested within Session. The Bonferroni correction was used to adjust p-values for multiple comparisons (Bland and Altman, 1995). Where applicable, both the uncorrected p-value and corrected p-value are presented for ease of interpretation since the Bonferroni correction increases the risk of Type II errors (Garamszegi, 2006). Linear regression models (R Core Team 2021) were used to investigate the association between the mean HR during the pre-stimulation resting period and treatment outcome, the association between the mean HR during the 1Hz rTMS period and treatment outcome, as well as the association between the reduction in HR from the pre-stimulation resting period to the 1Hz rTMS period and treatment outcome.

3. Results

3.1. Demographics and clinical outcome

24 participants were included in the analysis. The mean age in years of the study population was 41.3 ± 12.7 SD. Average baseline BDI-II was 33.2 ± 9.5 SD. Full demographic and clinical characteristics are given in Table 1. The response and remission rates at the second follow-up, which took place within seven days after the tapering sessions, were 35.4% and 27.1% respectively. Full clinical outcomes are summarized in Table 2.

3.2. Effect of 1Hz R-DLPFC rTMS on HR

To investigate the effect of 1Hz R-DLPFC rTMS on HR, HR during the pre-stimulation resting period was compared, using LMMs, to HR during each minute-long train of the 1Hz rTMS stimulation, as well as to HR during the post-stimulation resting period. Overall, 19 out of 25 patients (79.2%) showed a reduction in mean HR from the pre-stimulation resting period to the 1Hz rTMS period (Fig. 2c). Compared to the mean HR during the pre-stimulation resting period, the mean HR during the first minute of the 1Hz train was reduced by 2.87 bpm ($F(1, 3474) = 4.614$, $p = 0.0318$). HR also showed a reduction during the second train by 3.31 bpm and the third train by 3.02 bpm. These reductions showed significance as individual measurements, but not when corrected for multiple comparisons across 6 periods of stimulation ($F(1, 3447) = 6.31$, $p = 0.0121$ uncorrected, $p = 0.0722$ corrected), and 3.02 bpm ($F(1, 3453) = 5.39$, $p = 0.0203$ uncorrected, $p = 0.1217$ corrected). The mean HR during the subsequent stimulation periods at 4, 5, and 6 min were not statistically different from the baseline value and showed an increasing trend towards the baseline value (Fig. 1). The full statistical

Table 1
Demographic and clinical characteristics ($n = 24$).

Characteristics	No. (St.dev)
Age, years	41.3 (12.7)
Women	54.2%
Age of onset, years	24.3 (11.7)
Length of current depressive episode, months	55.3 (71.8)
Baseline BDI-II	33.2 (9.5)
Antidepressant	70.8%
Antidepressant combination	33.3%
Antipsychotic augmentation	25.0%
Psychostimulant augmentation	12.5%
Benzodiazepine	14.6%
ATHF total score	4.5 (4.3)
ATHF number of trials, current episode	1.3 (1.0)
ATHF highest score	2.7 (1.6)

Abbreviations: BDI-II = Beck Depression Inventory-II, ATHF = Antidepressant Treatment History Form.

Table 2
Clinical outcomes of 1Hz R-DLPFC arTMS ($n = 24$).

Outcome Measure: BDI-II	No. (St.dev)
Score at baseline	33.2 (9.5)
Score at second follow-up*	22.6 (13.4)
Mean percent improvement at second follow-up*	35.3% (35.1%)
Response rate (>50% improvement) at second follow-up*	33.3% (8/24)
Remission rate (≤12 on BDI-II) at second follow-up*	29.2% (7/24)

Abbreviations: BDI-II = Beck Depression Inventory-II.
* Second follow-up occurred ≤ 7 days after the last tapering session.

summary of the effects of 1Hz rTMS on HR is presented in Table 3.

3.3. HR and treatment outcome

Mean HR during the pre-stimulation resting period showed a non-significant trend toward a negative association with BDI-II percent improvement post treatment (p = 0.0836 uncorrected), with 13 percent of the variance in BDI-II percent improvement being explained by the model containing only the mean HR during the pre-stimulation resting period as the independent variable ($R^2 = 0.1299$) (Fig. 2a.).

Regarding HR during stimulation, since the second minute during the 1Hz R-DLPFC rTMS period was found to have the most significant within-subject reduction in HR from the pre-stimulation resting period, the mean HR during the second minute of the 1Hz R-DLPFC rTMS period was selected as the independent variable for subsequent linear regression models involving the 1Hz R-DLPFC rTMS period. First, we assessed the association between the mean HR during 1Hz R-DLPFC rTMS and treatment outcome. Then we tested for associations between the *reduction* in mean HR from the pre-stimulation resting period to the 1Hz rTMS period and treatment outcome. The mean HR during the 1Hz R-DLPFC rTMS period and the reduction in mean HR from the pre-stimulation resting period to the 1Hz R-DLPFC rTMS period showed a non-significant negative trend with treatment outcome (p = 0.2019 and 0.2983, $R^2 = 0.0729$ and 0.0490 respectively) (Fig. 2b and c).

4. Discussion

This study assessed the effects of 1Hz R-LPFC rTMS on HR among MDD patients, as well as the relationship between such effects and treatment outcome. In keeping with our previous results, a large proportion of patients, 79.2%, experienced a reduction in HR during the administration of 1Hz R-DLPFC-rTMS (Sheen et al., 2022). As

Table 3
Summary of the comparison of HR during the pre-stimulation resting session to HR during each minute of the 1Hz rTMS-EEG/ECG period and post-stimulation resting period. Statistical results from linear mixed effects models are reported.

Session	Mean HR, BPM (St. Error)	Statistical Comparisons (vs. Pre-Stimulation Resting)	Bonferroni Corrected P Value
Pre-Stimulation Resting	70.67 (0.57)	N/A	N/A
1Hz rTMS 1 min	67.85 (0.44)	$F = 4.61$, df = 1, df.res = 3474, $p = 0.0318$	0.1907
1Hz rTMS 2 min	67.36 (0.53)	$F = 6.31$, df=1, df.res = 3447, $p = 0.0121$	0.0722
1Hz rTMS 3 min	67.65 (0.45)	$F = 5.39$, df = 1, df.res = 3453, $p = 0.0203$	0.1217
1Hz rTMS 4 min	68.33 (0.61)	$F = 2.18$, df = 1, df.res = 3523, $p = 0.0880$	0.5280
1Hz rTMS 5 min	68.29 (0.60)	$F = 2.34$, df = 1, df.res = 3518, $p = 0.0790$	0.4740
1Hz rTMS 6 min	68.76 (0.58)	$F = 1.98$, df = 1, df.res = 3611, $p = 0.1050$	0.6300
Post-Stimulation Resting	69.16 (0.45)	$F = 0.47$, df = 1, df.res = 3839, $p = 0.4922$	1.0000

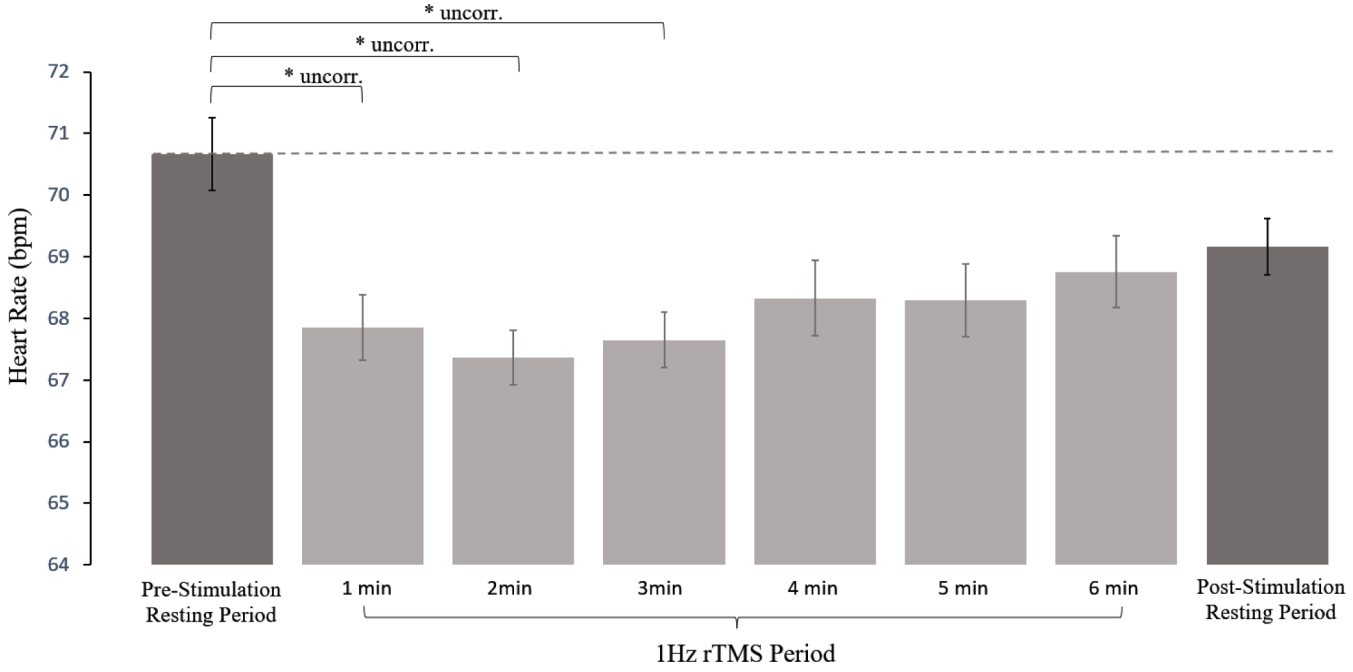


Fig. 1. Effect of 1Hz R-DLPFC rTMS on HR. First and last bars represent the mean HR over the pre-stimulation resting session and the post-stimulation resting session. Light gray bars represent the mean HR during each minute of the 6-minute 1Hz R-DLPFC rTMS session. Error bars represent standard errors of the mean. HR during the pre-stimulation resting period was compared to HR during each minute of the rTMS period, as well as to HR during the post-stimulation resting period. Compared to the mean HR during the pre-stimulation resting period, the mean HR during the first 3 min of the 1Hz R-DLPFC rTMS period was reduced by 2.87 bpm, 3.31 bpm, and 3.02 bpm respectively. * Uncorrected p-values: 1 min $p = 0.0318$, 2 min $p = 0.0121$, 3 min $p = 0.0203$; corrected: 1 min $p = 0.1907$, 2 min $p = 0.0722$, 3 min $p = 0.1217$. The dashed line represents the mean HR during the pre-stimulation resting period and demonstrates that, after an initial reduction, the mean HR of the 1Hz R-DLPFC rTMS period showed a trend towards returning to the baseline value.

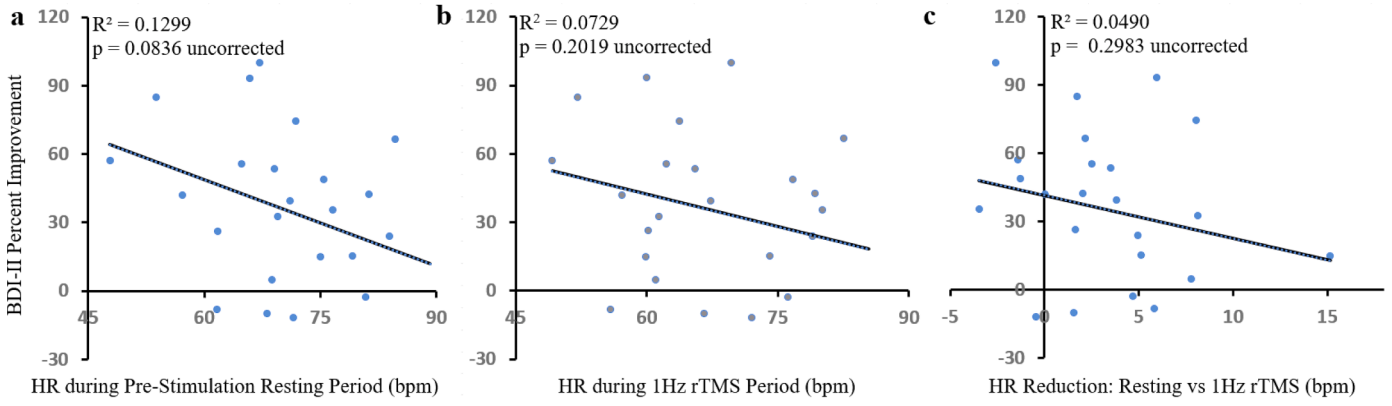


Fig. 2. Relationship between the mean heart rate and BDI-II percent improvement. Each dot represents one subject. **(a)** Linear regression showed a non-significant negative association between the mean heart rate during the pre-stimulation resting period and BDI-II percent improvement ($p = 0.0836$ uncorrected). 13 percent of the variance in BDI-II percent improvement was explained by the model containing only the mean heart rate during the pre-stimulation resting period as the independent variable ($R^2 = 0.1299$). **(b)** and **(c)** The mean HR during the second minute of the 1Hz R-DLPFC rTMS period and the reduction in mean HR from baseline to the second minute of the1Hz R-DLPFC rTMS period showed non-significant negative associations with treatment outcome ($p = 0.2019$ and 0.2983, $R^2 = 0.0729$ and 0.0490 respectively). **(c)** 19 out of 25 patients (79.2%) showed a reduction in mean HR from the pre-stimulation resting period to the second minute of the 1Hz rTMS period.

hypothesized, HR was, prior to the Bonferroni correction, significantly reduced from the pre-stimulation resting period to the first, second, and third minute of 1Hz R-DLPFC rTMS, in a manner similar to our previous results for 1Hz stimulation periods (Sheen et al., 2022), for ITIs (Voetterl et al., 2021), as well as the reductions observed for high-frequency or theta-burst rTMS of the left DLPFC (Iseger et al., 2020). However, the reductions observed in the current study did not reach statistical significance after the Bonferroni correction, while the reductions observed in our previous study remained significant even after the Bonferroni correction (Sheen et al., 2022). Consistent in trend with our previous reports for 1Hz stimulation periods (Sheen et al., 2022) and for ITIs (Voetterl et al., 2021), HR during the pre-treatment resting period, HR during 1Hz R-DLPFC rTMS period, and the reduction in HR from the pre-treatment resting period to the 1Hz R-DLPFC rTMS period all showed negative trends (non-significant) with treatment outcome in the present study.

As discussed more extensively in our previous study (Sheen et al., 2022), the observed reductions in HR during the rTMS period may be mediated through the interaction between the prefrontal cortex and the medullary nuclei onto which the branches of the vagus nerve converge (Park and Thayer, 2014). rTMS of the R-DLPFC may affect prefrontal input to the medulla and ultimately result in an increase in vagal tone and thereby increase parasympathetic activity. In MDD, patients have been shown to exhibit both an increase in resting HR and a decrease in heart rate variability (Michael and Kaur, 2021), which are posited to occur as the result of reduced parasympathetic activity, and thus correcting this parasympathetic deficiency may help to alleviate MDD symptoms. Notably, in both our present and previous study, the gradual return of HR during the rTMS period to near baseline levels after an pronounced initial reduction (Fig. 1.) is consistent with the timeframe of parasympathetic responses (Iseger et al., 2020). This suggests that parasympathetic activity may indeed be an important part of the underlying physiological mechanism of the observed results and that, in contrast to lengthy EEG and MRI sessions, only brief 3 min sessions may be necessary when utilizing ECG as a biomarker of rTMS treatment outcome. However, the negative trend between HR reduction and treatment outcome observed in this study, which is in concordance with the trends reported in our first study, contradictorily suggests that a larger parasympathetic activity increase (larger HR reduction) during rTMS will result in a worse treatment outcome. Thus, the hypothesized biological mechanism behind the observed HR reduction during the rTMS period should be further investigated in future studies.

The discrepancy in statistical significance between our previous reports and the present study could be due to the following differences in stimulation protocols: (i) differences in 1Hz duty cycle (60 s on, 30 s off x 6 trains in the present study, versus 600 pulses continuous in our previous study), (ii) differences in treatment schedule (6 times a day for 5 days with 20–25 once-daily tapering sessions in the present study, rather than 8 times a day for 5 days with no tapering in the previous study), and (iii) differences in the coils used (a ring-shaped coil centered on F4 in the present study, versus a figure-8 coil centered on F4 in the previous study). Any of the above factors could have contributed to the discrepancies in statistical significance.

Of particular relevance is the difference in coils. The ring-shaped coil used in the present study produces an electrical field which is weaker in the center (on F4) and stronger in the surrounding area, while the figure-8 coil shows the opposite pattern of a stronger field in the center and a weaker field over the surrounding area. Notably, both coils were able to elicit a bradycardia effect in the majority of patients. This observation is consistent with previous reports suggesting that bradycardia could be elicited at EEG sites adjacent to F4, such as FC4 (Iseger et al., 2017). At the same time, the proportion of responders and remitters was somewhat lower with ring-coil than with figure-8 coil stimulation (Sheen et al., 2022), despite a treatment schedule involving both more days and more overall sessions of stimulation. Future work using a randomized allocation will be needed to determine whether personalizing the DLPFC targets based on maximal neurocardiac effect can achieve significantly greater improvements in clinical symptoms.

This preliminary study has several limitations. First, due to the small sample size of this study, the previously described discrepancy in clinical outcome between our two studies may have simply risen from random variations in patient characteristics, since we did not have an opportunity to further investigate our hypothesis that the discrepancy is due to the differences in coils and targeting. This sample size is also too small to effectively control for possible confounding factors that may influence clinical outcome, such as medication regimen and treatment resistance (Kaster et al., 2019). Thus, importantly, future studies should consider having larger sample sizes to increase power and more effectively control for confounding factors. Second, the study made use of ECG recorded during EEG rather than on its own and did not control for possible effects of respiration on HR during recordings (Perry et al., 2019; Udupa et al., 2007; Udupa et al., 2011; Iseger et al., 2020). Finally, as an open-label study, the design did not incorporate a sham control as a baseline for interpretation of the effects of rTMS on either the HR or

clinical symptoms.

To conclude: the present study replicates and extends our previous finding that 1Hz rTMS of the right DLPFC elicits bradycardia in a majority of MDD patients (prior to the Bonferroni correction), showing this to be the case with a ring-shaped coil as was also the case for a figure-8 coil. The relationships between the various HR parameters and treatment outcome were weak and did not reach statistical significance in the present study; however, such results were expected due to the subpar clinical outcome of the ring coil, while, more importantly, the trends between HR and treatment outcome do agree with that of our previous report and thus support the utilization of ECG biomarkers for LF-rTMS treatment in MDD. Future work incorporating control arms with sham stimulation and comparing the effects of treatment at DLPFC targets with stronger versus weaker effects on HR may help to determine whether neurocardiac mapping is worth incorporating as a simple, scalable method for personalizing the rTMS target in individual patients for better treatment outcomes and higher chances of remission.

Role of the funding source

This work was supported in part by the Toronto General and Western Hospital Foundation and the Arrell Family Foundation.

CRediT authorship contribution statement

Jack Z. Sheen: Data curation, Formal analysis, Writing – original draft. **Jean-Philippe Miron:** Visualization, Writing – review & editing. **Helena Voetterl:** Visualization, Data curation, Writing – review & editing. **Farrokh Mansouri:** Formal analysis, Writing – review & editing. **Ryan Zhou:** Visualization, Data curation, Writing – review & editing. **Thomas Russell:** Visualization, Data curation, Writing – review & editing. **Molly Hyde:** Visualization, Data curation, Writing – review & editing. **Linsay Fox:** Visualization, Data curation, Writing – review & editing. **Zafiris J. Daskalakis:** Project administration, Writing – review & editing. **John D. Griffiths:** Project administration, Writing – review & editing. **Daniel M. Blumberger:** Project administration, Writing – review & editing. **Jonathan Downar:** Project administration, Writing – review & editing.

Declaration of Competing Interest

JZS, FM, KD, TR, RZ, MH, LF, HV do not report any conflict of interest.

DMB receives research support from the *Canadian Institutes of Health Research* (CIHR), National Institutes of Health – US (NIH), Brain Canada Foundation and the Temerty Family through the CAMH Foundation and the Campbell Family Research Institute. He received research support and in-kind equipment support for an investigator-initiated study from Brainsway Ltd. and he was the site principal investigator for three sponsor-initiated studies for Brainsway Ltd. He received in-kind equipment support from Magventure for investigator-initiated studies. He received medication supplies for an investigator-initiated trial from Indivior. He has participated in an advisory board for Janssen. He has participated in an advisory board for Welcony Inc.

ZJD has received research and equipment in-kind support for an investigator-initiated study through Brainsway Inc and Magventure Inc. His-work has been supported by the Canadian Institutes of Health Research (CIHR), the National Institutes of Mental Health (NIMH), Brain Canada and the Temerty Family and Grant Family and through the center for Addiction and Mental Health (CAMH) Foundation and the Campbell Institute.

JD has received research support from the Arrell Family Foundation, the Buchan Family Foundation, Brain Canada, the Canadian Biomarker Integration Network in Depression, the Canadian Institutes of Health Research (CIHR), the Klarman Family Foundation, NIH, the Ontario Brain Institute, the Toronto General and Western Hospital Foundation, and the Weston Family Foundation; he has received travel stipends from Lundbeck and ANT Neuro; he has served as an advisor for BrainCheck, Restorative Brain Clinics, and TMS Neuro Solutions.

JPM reports research grants from the Brain & Behavior Research Foundation (BBRF) Young Investigator Award and the Réseau Québécois sur le Suicide, les troubles de l'Humeur et les troubles Associés (RQSHA) as well as salary support for his graduate studies from the Branch Out Neurological Foundation.

JDG has received research support from the Labatt Family Network, the CAMH Discovery Fund, the University of Toronto EMHSeed program, and the Tri-Council Canada-UK AI Initiative.

Acknowledgments

We would like to sincerely thank Jane Lee, Jianjun Shen, Terri Cairo, Julian Kwok, Meaghan Todd, Nuno Ferreira, Igbinosa Uwadiae, Michael Aiello, Aisha Dar, Umar Dar, Vanathy Niranjan, Duong Nguyen, Sandy Ma, Peter Fettes, Laura Schulze, Arsalan Moghtadaei, Sunny Hong, Ekaterina Kosyachkova, and Eileen Lam for their most valued support, without which this study would not have been possible. We would also like to express our heartfelt appreciation for the trust, patience, and dedication of all the participants in this study.

References

Ahern, G.L., Sollers, J.J., Lane, R.D., et al., 2001. Heart rate and heart rate variability changes in the intracarotid sodium amobarbital test. Epilepsia 42 (7), 912–921. https://doi.org/10.1046/j.1528-1157.2001.042007912.x.

Bates, D., Mächler, M., Bolker, B., Walker, S., 2015. Fitting linear mixed-effects models using lme4. J. Stat. Softw. 67 (1), 1–48. https://doi.org/10.18637/jss.v067.i01.

Bland, J.M., Altman, D.G., 1995. Multiple significance tests: the Bonferroni method. BMJ 310 (6973), 170. https://doi.org/10.1136/bmj.310.6973.170.

Bagby, R.M., Ryder, A.G., Cristi, C., 2002. Psychosocial and clinical predictors of response to pharmacotherapy for depression. J. Psychiatry Neurosci. 27 (4), 250–257. PMID: 12174734.

Brunoni, A.R., Chaimani, A., Moffa, A.H., et al., 2017. Repetitive transcranial magnetic stimulation for the acute treatment of major depressive episodes: a systematic review with network meta-analysis. JAMA Psychiatry 74 (2), 143–152. https://doi.org/10.1001/jamapsychiatry.2016.3644.

Casanova, M.F., Hensley, M.K., Sokhadze, E.M., et al., 2014. Effects of weekly low-frequency rTMS on autonomic measures in children with autism spectrum disorder. Front. Hum. Neurosci. https://doi.org/10.3389/fnhum.2014.00851, 0.

Friedrich, M.J., 2017. Depression is the leading cause of disability around the world. JAMA 317 (15), 1517. https://doi.org/10.1001/jama.2017.3826, 1517.

Garamszegi, L.Z., 2006. Comparing effect sizes across variables: generalization without the need for Bonferroni correction. Behav. Ecol. 17 (4), 682–687. https://doi.org/10.1093/beheco/ark005.

Iseger, T.A., Arns, M., Downar, J., Blumberger, D.M., Daskalakis, Z.J., Vila-Rodriguez, F., 2020. Cardiovascular differences between sham and active iTBS related to treatment response in MDD. Brain Stimul. Basic Transl. Clin. Res. Neuromodulation 13 (1), 167–174. https://doi.org/10.1016/j.brs.2019.09.016.

Iseger, T.A., Padberg, F., Kenemans, J.L., Gevirtz, R., Arns, M., 2017. Neuro-cardiac-guided TMS (NCG-TMS): probing DLPFC-sgACC-vagus nerve connectivity using heart rate - first results. Brain Stimul. 10 (5), 1006–1008. https://doi.org/10.1016/j.brs.2017.05.002.

Kaur, M., Michael, J.A., Hoy, K.E., et al., 2020. Investigating high- and low-frequency neuro-cardiac-guided TMS for probing the frontal vagal pathway. Brain Stimul. Basic Transl. Clin. Res. Neuromodulation 13 (3), 931–938. https://doi.org/10.1016/j.brs.2020.03.002.

Kaster, T.S., Downar, J., Vila-Rodriguez, F., et al., 2019. Trajectories of response to dorsolateral prefrontal rTMS in major depression: a THREE-D study. Am. J. Psychiatry 176 (5), 367–375. https://doi.org/10.1176/appi.ajp.2018.18091096.

Lefaucheur, J.P., Aleman, A., Baeken, C., et al., 2020. Evidence-based guidelines on the therapeutic use of repetitive transcranial magnetic stimulation (rTMS): an update (2014–2018). Clin. Neurophysiol. 131 (2), 474–528. https://doi.org/10.1016/j.clinph.2019.11.002.

Miron, J.P., Sheen, J., Panzarella, T., et al., 2021b. The relationship between pre-treatment heart rate variability and response to low-frequency accelerated repetitive transcranial magnetic stimulation in major depression. J. Affect. Disord. Rep. 6, 100270 https://doi.org/10.1016/j.jadr.2021.100270.

Miron, J.P., Sheen, J., Mansouri, F., et al., 2020. The role of low-frequency repetitive transcranial magnetic stimulation in major depression: a call to increase the evidence base. Brain Stimul. Basic Transl. Clin. Res. Neuromodulation 13 (5), 1296–1297. https://doi.org/10.1016/j.brs.2020.06.013.

McClintock, S.M., Reti, I.M., Carpenter, L.L., et al., 2017. Consensus recommendations for the clinical application of repetitive transcranial magnetic stimulation (rTMS) in the treatment of depression. J. Clin. Psychiatry 78 (1). https://doi.org/10.4088/JCP.16cs10905, 0-0.

Mir-Moghtadaei, A., Dunlop, K., Mansouri, F., et al., 2017. Scalp-based heuristics for locating the nodes of the salience network for use in neurostimulation. Brain Stimul. Basic Transl. Clin. Res. Neuromodulation 10 (2), 479. https://doi.org/10.1016/j.brs.2017.01.404.

Milev, R.V., Giacobbe, P., Kennedy, S.H., et al., 2016. Can. J. Psychiatry 61 (9), 561–575. https://doi.org/10.1177/0706743716660033.

Makovac, E., Thayer, J.F., Ottaviani, C., 2017. A meta-analysis of non-invasive brain stimulation and autonomic functioning: implications for brain-heart pathways to cardiovascular disease. Neurosci. Biobehav. Rev. 74, 330–341. https://doi.org/10.1016/j.neubiorev.2016.05.001.

Michael, J.A., Kaur, M., 2021. The heart-brain connection in depression: can it inform a personalised approach for repetitive transcranial magnetic stimulation (rTMS) treatment? Neurosci. Biobehav. Rev. 127, 136–143. https://doi.org/10.1016/j.neubiorev.2021.04.016.

Miron, J.P., Voetterl, H., Fox, L., et al., 2021a. Optimized repetitive transcranial magnetic stimulation techniques for the treatment of major depression: a proof of concept study. Psychiatry Res. 298, 113790 https://doi.org/10.1016/j.psychres.2021.113790.

Park, G., Thayer, J.F., 2014. From the heart to the mind: cardiac vagal tone modulates top-down and bottom-up visual perception and attention to emotional stimuli. Front Psychol. 5, 278. https://doi.org/10.3389/fpsyg.2014.00278.

Plotly Technologies Inc, 2015. Collaborative data science. Montréal, QC. Plotly Technologies Inc. https://plot.ly.

Perry, S., Khovanova, N.A., Khovanov, I.A., 2019. Control of heart rate through guided high-rate breathing. Sci. Rep. 9 (1), 1545. https://doi.org/10.1038/s41598-018-38058-5.

Rossi, S., Hallett, M., Rossini, P.M., Pascual-Leone, A., Safety of TMS Consensus Group, 2009. Safety, ethical considerations, and application guidelines for the use of transcranial magnetic stimulation in clinical practice and research. Clin. Neurophysiol. 120 (12), 2008–2039. https://doi.org/10.1016/j.clinph.2009.08.016.

R Core Team, 2021. R: A language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria. https://www.R-project.org/.

Sheen, J.Z., Miron, J.P., Mansouri, F., et al., 2022. Cardiovascular biomarkers of response to accelerated low frequency repetitive transcranial magnetic stimulation in major depression. J. Affect. Disord. 318, 167–174. https://doi.org/10.1016/j.jad.2022.08.105.

Simon, G.E., Perlis, R.H., 2010. Personalized medicine for depression: can we match patients with treatments? Am. J. Psychiatry 167 (12), 1445–1455. https://doi.org/10.1176/appi.ajp.2010.09111680.

Udupa, K., Sathyaprabha, T.N., Thirthalli, J., Kishore, K.R., Raju, T.R., Gangadhar, B.N, 2007. Modulation of cardiac autonomic functions in patients with major depression treated with repetitive transcranial magnetic stimulation. J. Affect. Disord. 104 (1), 231–236. https://doi.org/10.1016/j.jad.2007.04.002.

Udupa, K., Thirthalli, J., Sathyaprabha, T.N., Kishore, K.R., Raju, T.R., Gangadhar, B.N., 2011. Differential actions of antidepressant treatments on cardiac autonomic alterations in depression: a prospective comparison. Asian J. Psychiatry 4 (2), 100–106. https://doi.org/10.1016/j.ajp.2011.02.006.

Voetterl, H., Miron, J.P., Mansouri, F., et al., 2021. Investigating EEG biomarkers of clinical response to low frequency rTMS in depression. J. Affect. Disord. Rep. 6, 100250 https://doi.org/10.1016/j.jadr.2021.100250.